Frequency Scaling, Band Edges, and the Role of the Phononic Gap

Optomechanics project (SnV cavity QED) — discussion notes, 2026-10-14. Rough, order-of-magnitude derivations; items marked "to verify" are estimates not taken from the papers.

1. Optical-to-mechanical frequency ratio f_o/f_m

Empirically, each OMC geometry class has a roughly fixed ratio between its optical and mechanical frequencies. Values from the papers in *docs/papers*:

Device	Material	f_o	f_m	f_o/f_m
Chan 2012, 1D nanobeam, breathing	Si	194 THz	5.1 GHz	3.8×10^4
Joe 2026, 1D rectangular beam, breathing	diamond	409.7 THz	12.06 GHz	3.4×10^4
Meesala Ch. 6, triangular beam, flapping	diamond	192 THz	4.9 GHz	3.9×10^4
Meesala Ch. 6, triangular beam, swelling	diamond	192 THz	8.1 GHz	2.4×10^4
Safavi-Naeini 2014, 2D snowflake	Si	195 THz	9.3 GHz	2.1×10^4
Mayor 2025, 2D b-dagger	Si	191.7 THz	7.44 GHz	2.6×10^4

Optical side (same for 1D and 2D)

The cavity mode is built from a band-edge state at the zone boundary, $k = \pi/a$ (Section 2). There the field is a standing wave with period 2a, i.e. $\lambda/n_{eff} \approx 2a$:

$$f_o \approx c / (2\, n_{eff}\, a) \text{ i.e. } a/\lambda \approx 1/(2\, n_{eff})$$

Check (Chan 2012, defect cell): a = 327 nm, f_o = 194 THz → $n_{eff} \approx 2.4$, below bulk Si (3.48) because of the holes and vertical confinement.

2D snowflake: mechanical gap tied to the period

The phononic gap sits at $f_m = \eta\, v/a$ (Section 3). Dividing:

$$f_o/f_m \approx [(a/\lambda) / \eta] \times (c/v)$$

The period cancels; only the dimensionless band positions matter.

- **Si (S&P 2014):** photonic gap at $a/\lambda \approx 0.325$, phononic $\eta \approx 0.70$ (v_{TA} = 5860 m/s) → $0.46 \times 3\times10^8/5860 \approx 2.4\times10^4$ (measured 2.1×10^4; the cavity mode sits above gap center).
- **Diamond (our sweeps):** $f\cdot a \approx 8000$ GHz·nm → $\eta \approx 0.68$ (v_{TA} = 11,840 m/s), so $f_o/f_m \approx (a/\lambda) \times 3.7\times10^4$. The unknown is a/λ of the diamond snowflake photonic gap; lower index pushes it up, plausibly 0.35–0.45 → ratio $1.3\text{–}1.7\times10^4$ → **$f_m \approx$ 28–37 GHz at 619 nm**. The diamond optical band structure (pending OP study) pins this down.

1D nanobeam: breathing mode tied to the beam width

The Γ-point breathing mode is a transverse standing wave across the beam (a local resonance of each cell), not a Bragg mode along it. With width w and a softening factor $\beta < 1$ from the hole:

$$f_m \approx \beta\, v/(2w) \rightarrow f_o/f_m \approx [c/(n_{eff}\, v)] \times (w/a) / \beta$$

Larger than 2D because (i) w > a, so the transverse acoustic wavelength exceeds the period, and (ii) the fins beside the hole are soft (no connected 2D network). Check (Chan 2012): $c/(n_{eff} v_{LA}) = 3\times10^8/(2.36 \times 8433) \approx 1.5\times10^4$; × w/a = 1.6, ÷ β ≈ 0.65 → 3.7×10^4 (measured 3.8×10^4).

Correction to an earlier claim: scaling Si → diamond by c/(n v) alone predicts the diamond ratio should be ~0.7× Si; the 1D data show ~0.9×. Hypothesis (to verify against Joe 2026 supplementary dimensions, not in docs/papers): at n = 2.41 the beam must be wider relative to its period to guide the mode, so w/a compensates. The 2D diamond estimate above uses our own simulated η rather than this scaling.

Implication at 619 nm ($f_o \approx 484$ THz)

- 2D single-scale (snowflake): f_m tied to a, which is fixed by the photonic gap → ~30 GHz.
- 1D breathing mode: f_m set by w, partly independent of a → 484 THz / $3.4\times10^4 \approx$ **14 GHz**. Lower requires wider beams or softer breathing modes, at a cost in optical confinement or g_0.
- Mayor 2025 states the same trend: fully connected 2D structures are stiffer (>10 GHz) than 1D (<5 GHz in Si). A ≤10 GHz target with a single-scale 2D design at 619 nm conflicts with this physics.

2. Why the cavity mode sits near the band edge at k = π/a

Bragg condition

A periodic ε(x) has Fourier components at G = 2π/a and couples a wave at k to one at k − G. The coupling is resonant when both have the same frequency, |k| = |k − G|, i.e. k = G/2 = π/a (equivalently 2a = λ/n_{eff}). Forward and backward waves mix into two standing waves of period 2a: one concentrated in high-ε material (lower frequency, *dielectric band*) and one in the holes (*air band*). Their splitting is the bandgap, so band edges are at the zone boundary (X for a nanobeam; X of the line-defect waveguide for S&P 2014).

Effective-mass (envelope) picture

At a band edge the group velocity is zero and the band is locally parabolic:

$$\omega(k) \approx \omega_{edge} + \alpha\,(k - \pi/a)^2$$

A local perturbation that shifts the band edge into the mirror's gap (Chan 2012: smaller a and hole aspect ratio raise the X-point dielectric band) acts as a potential well and binds a state:

$$E(x) \approx u_{\pi/a}(x) \cdot F(x)$$

$u_{\pi/a}$ is the band-edge Bloch function (fast pattern) and F(x) a slow, roughly Gaussian envelope. Because F is smooth, the mode's Fourier content is concentrated near k = π/a and its frequency lies just inside the gap.

Why X and not Γ (slab loss)

In a slab, Fourier components with $|k_{||}| < \omega/c$ lie inside the light cone and radiate. Γ-point states (k ≈ 0) are inside it; X-point states (k = π/a) are outside it if ω is below the light line there (S&P 2014 emphasize their gap lies well below the light line at the zone boundary). A cavity made of X-point states with a smooth envelope has little weight near k ≈ 0, hence high Q. This is why designs use gentle tapers (quadratic a-profile in Chan 2012, 3% quadratic r-variation in S&P 2014); abrupt defects put Fourier tails into the light cone.

The mechanical breathing mode uses Γ instead: phonons have no light cone into vacuum and the breathing band has an extremum at Γ, so the same envelope argument applies there. (pccsim heterostructure runs evaluate only k = 0 because the supercell contains the whole localized mode, which barely depends on the supercell Bloch vector; it is still built from waveguide states near k = π/a.)

3. Meaning of the geometry factor η in $f_{gap} = \eta\, v/a$

For elastic waves the Bragg gap also opens at k = π/a, i.e. phonon wavelength 2a ("half a wavelength per period"):

$$f_{gap} \approx v/(2a) \text{ i.e. } f\cdot a/v \approx 0.5 = \eta \text{ (ideal 1D Bragg mirror)}$$

Writing f·a/v = η gives one dimensionless number per geometry, comparable across materials and lattice constants (S&P 2014: 0.70; our Si validation: 0.582). Reasons the real η differs from 0.5:

- **Which velocity:** a slab carries longitudinal (~17.5 km/s in diamond), transverse (~12 km/s) and dispersive flexural modes ($\omega \sim k^2$ at long wavelength). Quoting against v_{TA} is a convention: the same gap is $\eta \approx 0.68$ with v_{TA} or 0.46 with v_{LA}.
- **Holes soften the medium:** the effective velocity of the patterned slab is below bulk.
- **2D zone boundaries differ:** triangular lattice M at $|k| = 2\pi/(\sqrt{3}\,a)$ ($\lambda = \sqrt{3}\,a$), K at $4\pi/(3a)$ ($\lambda = 1.5a$). A complete gap must be open in all directions at once.
- **Mainly, the snowflake gap is resonance-type, not Bragg:** S&P describe triangles joined by thin bridges. Triangle resonances give flat "optical-like" bands; bridge stiffness sets the "acoustic-like" bands. Narrow bridges pull the acoustic-like bands down, opening a gap analogous to the acoustic/optical phonon gap of an atomic crystal.

d/a (diamond_a_sweep)	f·a (GHz·nm)	$\eta = f \cdot a / v_{TA}$
0.35	≈ 8900	≈ 0.75
0.41	≈ 8000	≈ 0.68
0.48	≈ 6500–7400	≈ 0.55–0.63

Longer arms leave narrower bridges → softer acoustic-like bands → lower and wider gap, consistent with the resonance picture. f·a also drifts with a at fixed ratios (a pure Bragg gap would scale exactly as 1/a) because t = 140 nm and the 16 nm fillet do not scale. diamond_a_large (t/a down to ~0.14) tests this hardest: flexural bending of a thin slab can soften the bridges much more than at small a.

4. What the phononic gap is for: radiation shield vs. bath suppression

- **Radiation shield** (Chan 2012, Joe 2026 outlook, current CLAUDE.md Stage 2 text): a gap around a *mechanical mode* stops leakage to the substrate and raises Q_m. Gap frequency set by the mode (~10–15 GHz).
- **Bath suppression** (Kuruma 2025; the original intent, as in agents/codesign_mediator.md): a gap at the *emitter's* phonon-coupled transition removes the density of states driving relaxation/dephasing. Gap frequency set by the emitter.

Where SnV phonon-driven decoherence sits

Process	Phonon frequency	Limits	Reachable by a PnC?
Single-phonon orbital relaxation, ground state	~850 GHz (Δ_{gs})	Spin T_2 above ~2–4 K (rate ~ $e^{-\Delta/kT} \times$ large prefactor)	No (a ≈ 9 nm)
Two-phonon Raman	Broadband to ~kT/h (~80 GHz at 4 K; thermal peak ~230 GHz)	Orbital T_1 at higher T	Only a small fraction
Excited-state orbital / ZPL broadening	~3 THz (Δ_{es})	Optical linewidth	No
Phonon sideband (DWF 0.57)	Local vibrations, THz	ZPL fraction	No
Direct spin flip	~5–40 GHz (Zeeman, set by B)	Spin T_1 at mK	Yes, but likely not the bottleneck

Required lattice constant $a \approx \eta\, v_{TA}/f$ with $\eta \approx 0.65$, $v_{TA} \approx 12$ km/s: 60 GHz → 130 nm (Kuruma, 20 nm features); 200 GHz → ~40 nm (sub-10 nm features); 850 GHz → ~9 nm. The large spin-orbit splitting that makes SnV

coherent at ~1.7 K also puts its phonon-driven decoherence beyond any fabricable PnC. Kuruma works because SiV's $\Delta_{gs} \approx 50$ GHz is reachable; their key control showed plain nanobeams (no complete gap) give no $T_{1,orbit}$ improvement.

> CLAUDE.md says 850 GHz processes are "thermally frozen out below ~40 K". hf/k ≈ 41 K, so they are exponentially suppressed, not frozen out; SnV coherence still degrades noticeably above a few kelvin.

SnV spin–phonon coupling (to verify)

In Joe 2026, g_{sm} scales with B_{perp} ($\sin^2\theta$ fit of the Purcell rate): the spin couples to strain only through spin-orbit mixing, roughly as $\gamma B_{perp}/\Delta_{gs}$. With $\Delta_{gs} \approx 850$ GHz vs ~50 GHz for SiV, SnV spin-strain susceptibility, and g_{sm}, should be ~10× lower at equal field, ~100× in cooperativity. This is an estimate, not stated in the papers; check the SnV strain literature before fixing the Stage 3 target.

Options

- Target spin T_1 at the Zeeman frequency (10–40 GHz overlaps the snowflake results); gain may be small for SnV.
- Partial Raman suppression; modest at 4 K but computable from the gapped vs. ungapped phonon DOS.
- Switch to SiV: $\Delta_{gs} \approx 50$ GHz matches snowflakes at a ≈ 150–200 nm (~40–50 GHz), but 737 nm photonics wants a ≈ 300 nm (mismatch ~1.7×).
- Keep SnV, use the gap as a radiation shield (Stage 3), and rely on ≤1.7 K operation for the bath.

Open decision: SnV specifically, or bath suppression as the core idea (possibly with another emitter)? CLAUDE.md and codesign_mediator.md should be reconciled once decided.

5. Paper takeaways (brief)

- **Safavi-Naeini 2014 (PRL):** Si snowflake, (d, r, w, a) = (220, 200–210, 75, 500) nm (text says r = 210, Fig. 1 caption 200). Phononic gap 6.9–9.5 GHz; optical pseudo-gap 185–235 THz below the light line at X. Cavity by 3% quadratic r-variation of a line-defect waveguide; $f_m = 9.3$ GHz, $g_0/2\pi = 220$ kHz, $Q_{o,i} = 1.8\times10^5$. Flat acoustic waveguide dispersion makes mechanical modes disorder-sensitive.
- **Chan 2012 (APL):** Si nanobeam, nominal (a, t, w, h_x, h_y) = (436, 220, 529, 165, 366) nm. Nelder–Mead optimization of $g\cdot\min(Q, Q_{cutoff})$. Simulated g = 860 kHz = +950 kHz photoelastic − 90 kHz moving boundary; measured 1.1 MHz, $Q_m = 6.8\times10^5$ at 10 K. Cross shield (534, 454, 134, 220) nm centred on 5.1 GHz.
- **Mayor 2025 (Nat. Commun.):** Si 2D b-dagger, a = 448 nm; $f_m = 7.44$ GHz, $g_0/2\pi \approx 880$–920 kHz, $Q_o = 2.4\times10^5$. 8× better thermalization than 1D; ground-state cooling from 3 K; strong coupling ($4g/\kappa \approx 1.17$).
- **Kuruma 2025 (Nat. Phys.):** diamond block–tether 1D PnC, a = 130 nm, t ≈ 22 nm, slab 70 nm; complete gap 50–70 GHz. SiV $T_{1,orbit}$ up to 18× bulk; a complete gap is required; two-phonon Raman (T^5) takes over above 12 K.
- **Joe 2026 (Nature):** diamond 1D OMC, 409.7 THz, $Q_o \approx 27{,}000$, $f_m = 12.06$ GHz, $Q_m \approx 34{,}000$ (4 K). $C_o = 31$; acoustic Purcell 10×, $g_{sm}/2\pi \approx 300$ kHz (9 MHz ideal). Cavity tuning broadened the mechanical line from 350 kHz to ~200 MHz (gas) or ~35 MHz (ALD + gas), limiting cooperativity; outlook names phononic-bandgap shielding as the next step.
- **Meesala 2019 thesis, Ch. 6:** diamond OMCs at 4 K: triangular cross-section flapping mode 4.9 GHz with $Q_m = 2.2\times10^5$; isotropically etched rectangular beams with 9.1 GHz breathing mode, $Q_m \approx 36{,}000$. SiV strain susceptibility ~100 THz/strain.